Adrien Allorant

Lecturer B in Social Statistics and Global Health
University of Southampton

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Department of Social Statistics and Demography
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Academic Appointments

University of Southampton 2024–present
Lecturer B in Social Statistics and Global Health, Department of Social Statistics and Demography

- Deputy Director, Southampton Statistical Sciences Research Institute (S3RI) 2026–present
- Undergraduate Studies Lead, Department of Social Statistics and Demography
- Personal Academic Tutor 2024–present

Reed College 2023–2024
Visiting Assistant Professor in Statistics, Department of Mathematics and Statistics

McGill University and Imperial College London 2022–2023
Postdoctoral Research Fellow, Department of Epidemiology and Biostatistics, McGill University; advisers Mathieu Maheu-Giroux and Jeffrey W. Imai-Eaton

- Led the UNAIDS Granular Data Analysis for Eastern and Southern Africa: 18 country reports, with three PhD students.
- Worked on Shiny90 and Naomi, the models sponsored by UNAIDS for the official HIV estimates of more than 40 countries.
- Facilitated UNAIDS regional workshops on HIV estimates with national programmes.

University of Washington, Institute for Health Metrics and Evaluation 2017–2021
Graduate Research Associate

- Small area estimation for health facility surveys (Local Burden of Disease project).
- Joint modelling of tuberculosis prevalence and notification data in Bangladesh (with icddr,b).
- COVID-19 modelling response team: global excess mortality, infections and infection-fatality ratios.

Education

PhD in Global Health Metrics, University of Washington 2021
Dissertation: Measuring inequities in health service delivery and their impact on health outcomes in low- and middle-income countries. Chair: Robert C. Reiner Jr.
Certificate in Statistics, Center for Statistics and the Social Sciences

Diplôme d'Ingénieur Statisticien-Économètre, ENSAE Paris 2016
MSc in Statistics and Econometrics, statistical learning track; with distinction

MPH in Public Health Research, Paris Saclay University 2016
With distinction; Best Master's Thesis Award

MRes in Sociology, Sciences Po Paris 2015

BSc in Applied Mathematics, Université Paris I Panthéon-Sorbonne 2013
Concentration in probability and statistics; with high honours

BA in Social Sciences, Sciences Po Paris 2013
Concentration in economics; cum laude

Research

Computational and quantitative methods for the production of official and social statistics: calibration of administrative data with small probability samples; survey design and small area estimation; artificial intelligence and machine learning in national statistical offices; measurement of socio-economic status and poverty; population health in African countries.

Publications

Peer-reviewed articles

Allorant A, Bergen N, Danovaro-Holliday MC, Lorin J, Caetano Corrêa G, Boyda D, Belanger JL, Santhana Gopala Krishnan RS, Panciera R, Hosseinpoor AR. 2026. "Assessing geographic inequalities in childhood immunisation coverage: a critical scoping review of spatial analysis methods." *Vaccines* 14(7): 572. doi:10.3390/vaccines14070572

Allorant A, Smith PA. 2026. "Algorithm-assisted inference and the future of official statistics." *Journal of Official Statistics*. doi:10.1177/0282423X261443590

Allorant A, Bekelynck A, Monroe-Wise A, Baptista da Silva C, Chidarikire T, Edun O, Joaquim L, Kisia C, Larmarange J, Lyimo JJ, James JM, Musanhu C, Ncube G, Sathane I, Simo-Fotso A, Taasi G, Johnson CC. 2026. "Scaling up HIV self-testing in Africa: insights from national programmatic data in eight countries." *Journal of the International AIDS Society* 29(5): e70120. doi:10.1002/jia2.70120

Allorant A, Kuchukhidze S, Stannah J, Xia Y, Masuku SS, Ekanmian GK, Imai-Eaton JW, Maheu-Giroux M. 2025. "Socio-demographic and geographic disparities in HIV prevalence, HIV testing and treatment coverage: an analysis of 108 national household surveys in 33 African countries." *Journal of the International AIDS Society* 28(8): e70024. doi:10.1002/jia2.70024

Lyons C, Nambiar D, Johns NE, **Allorant A**, Bergen N, Hosseinpoor AR. 2024. "Inequality in childhood immunization coverage: a scoping review of data sources, analyses, and reporting methods." *Vaccines* 12(8): 850.

Johns NE, Blumenberg C, Kirkby K, **Allorant A**, Dos Santos Costa F, Danovaro-Holliday MC, Lyons C, Yusuf N, Barros AJD, Hosseinpoor AR. 2024. "Comparison of wealth-related inequality in tetanus vaccination coverage before and during pregnancy: a cross-sectional analysis of 72 low- and middle-income countries." *Vaccines* 12(4): 431.

Allorant A, Muset P, Hodgins C, Kirakoya-Samadoulougou F, Namachapa K, Mbofana F, Panagiotoglou D, Johnson LF, Imai-Eaton JW, Maheu-Giroux M. 2024. "Temporal trends and determinants of HIV testing at antenatal care in Sub-Saharan Africa: a pooled analysis of population-based surveys (2005–2021)." *JAIDS Journal of Acquired Immune Deficiency Syndromes* 95(1S): e97–e105.

Allorant A, Fullman N, Leslie HH, Sarr M, Gueye D, Eliakimu E, Wakefield J, Dieleman JL, Pigott D, Puttkammer N. 2023. "A small area model to assess temporal trends and sub-national disparities in healthcare quality." *Nature Communications* 14: 4555. doi:10.1038/s41467-023-40234-9

Barber RM, Sorensen RJD, Pigott DM, Bisignano C, Carter A, Amlag JO, Collins JK, Abbafati C, Adolph C, **Allorant A**, et al. 2022. "Estimating global, regional, and national daily and cumulative infections with SARS-CoV-2 through Nov 14, 2021: a statistical analysis." *The Lancet* 399(10344): 2351–2380.

Sorensen RJD, Barber RM, Pigott DM, Carter A, Spencer CN, Ostroff SM, Reiner RC, Abbafati C, Adolph C, **Allorant A**, et al. 2022. "Variation in the COVID-19 infection-fatality ratio by age, time, and geography during the pre-vaccine era: a systematic analysis." *The Lancet* 399(10334): 1469–1488.

Bollyky TJ, Hulland EN, Barber RM, Collins JK, Kiernan S, Moses M, Pigott DM, Reiner RC Jr, Sorensen RJD, Abbafati C, **Allorant A**, et al. 2022. "Pandemic preparedness and COVID-19: an exploratory analysis of infection and fatality rates, and contextual factors associated with preparedness in 177 countries." *The Lancet* 399(10334): 1489–1512.

Wang H, Paulson KR, Pease SA, Watson S, Comfort H, Zheng P, Aravkin AY, Bisignano C, Barber RM, Alam T, **Allorant A**, et al. 2022. "Estimating excess mortality due to the COVID-19 pandemic: a systematic analysis of COVID-19-related mortality, 2020–21." *The Lancet* 399(10334): 1513–1536.

Allorant A, Biswas S, Ahmed S, Wiens KE, LeGrand KE, Janko MM, Henry NJ, Dangel WJ, Watson A, Blacker BF, et al. 2022. "Finding gaps in routine tuberculosis surveillance activities in Bangladesh." *The International Journal of Tuberculosis and Lung Disease* 26(4): 356.

Celestin K, **Allorant A**, Virgin M, Marinho E, Francois K, Honoré JG, White C, Valles JS, Perrin G, De Kerorguen N, et al. 2021. "Short-term effects of the COVID-19 pandemic on HIV care utilization, service delivery, and continuity of HIV antiretroviral treatment in Haiti." *AIDS and Behavior* 25(5): 1366–1372.

Allorant A, Parrish C, Desforges G, Robin E, Honoré JG, Puttkammer N. 2020. "Closing the gap in implementation of HIV clinical guidelines in a low resource setting using electronic medical records." *BMC Health Services Research* 20(1): 804.

Naudet J, **Allorant A**, Ferry M. 2018. "Heirs, corporate aristocrats and 'meritocrats': the social space of top CEOs and Chairmen in India." *Socio-Economic Review* 16(2): 307–339.

Naudet J, **Allorant A**, Ferry M. 2018. "Héritiers, aristocrates d'entreprise et 'méritocrates': l'espace social du grand patronat en Inde." *Entreprises et Histoire* 90(1): 88–105.

Prud'Homme C, Deschamps F, **Allorant A**, Massard C, Hollebecque A, Yevich S, Ngo-Camus M, Gravel G, Nicotra C, Michiels S, et al. 2018. "Image-guided tumour biopsies in a prospective molecular triage study (MOSCATO-01): what are the real risks?." *European Journal of Cancer* 103: 108–119.

Caramella C, **Allorant A**, Orlhac F, Bidault F, Asselain B, Ammari S, Jaranowski P, Moussier A, Balleyguier C, Lassau N. 2018. "Can we trust the calculation of texture indices of CT images? A phantom study." *Medical Physics* 45(4): 1529–1536.

Verlingue L, Malka D, **Allorant A**, Massard C, Ferté C, Lacroix L, Rouleau E, Auger N, Ngo M, Nicotra C, et al. 2017. "Precision medicine for patients with advanced biliary tract cancers: an effective strategy within the prospective MOSCATO-01 trial." *European Journal of Cancer* 87: 122–130.

Harttrampf AC, Lacroix L, Deloger M, Deschamps F, Puget S, Auger N, Vielh P, Varlet P, Balogh Z, Abbou S, **Allorant A**, et al. 2017. "Molecular screening for cancer treatment optimization (MOSCATO-01) in pediatric patients: a single-institutional prospective molecular stratification trial." *Clinical Cancer Research* 23(20): 6101–6112.

Book chapter

Naudet J, **Allorant A**, Ferry M. 2019. "Heirs, corporate aristocrats, and 'meritocrats': the social space of top CEOs and chairmen in India." In *Mapping the Elite: Power, Privilege, and Inequality*. Oxford University Press.

Accepted, under review and submitted

Allorant A, Amane G, Cumbana E, Magaia H, Mondlane J, Johnson C, Couto A, Gaspar I. "HIV testing services in Mozambique one year after the PEPFAR stop-work order." *BMJ Global Health*. Accepted with minor revisions.

Youssif O, **Allorant A**. "Mismeasuring the poorest? The wealth index and two decades of household welfare in Egypt, Jordan and Palestine." *PLOS ONE*. Under review.

Allorant A, Zhang L-C. 2026. "A statistical calibration framework for subnational health surveillance from routine data and survey audits." Submitted.

Work in progress

Allorant A. "A common measure of socio-economic status? Opening the black box of the DHS wealth index by analysing its metadata." In preparation.

Allorant A, Perfetti Villa L, Luna Hernandez A, Tzavidis N. "Mismatched measures: comparing asset-based and consumption-based poverty in Mozambique." In preparation.

Other publications and reports

Allorant A. 2026. Contribution to the Discussion of "Some statistical aspects of the COVID-19 response" by Wood et al. *Journal of the Royal Statistical Society Series A: Statistics in Society* 189(1): 59–60.

Allorant A, Naudet J. 2019. "The politics of antibiotics: an interview with Ramanan Laxminarayan." *Books and Ideas*.

UNAIDS Regional Support Team for Eastern and Southern Africa. 2023. *Granular data analysis of HIV burden and engagement with care*: 18 country reports. Lead analyst.

Allorant A. 2021. *Measuring inequities in health service delivery and their impact on health outcomes in low- and middle-income countries*. PhD thesis, University of Washington.

Grants and Research Contracts

Review of survey sampling and weighting (£40,000), Scottish Government. Co-principal investigator — 2026–2027

Global Partnership Award, Africa (£9,016), University of Southampton. Sole principal investigator — 2026–2027

HIV self-testing analysis (US$25,000), World Health Organization. Agreement for Performance of Work — 2025–2026

Small Research Grant, HIV pilot evaluation in Mozambique (£10,000), British Academy and Leverhulme Trust. Principal investigator — 2025–2026

Pilot Award, AI in official statistics (£3,000), Web Science Institute, University of Southampton. Principal investigator — 2025

Research Culture Development Fund (£1,015), Faculty of Social Sciences, University of Southampton. Co-recipient with Professor Ann Berrington — 2024–2025

Spatial methods for childhood immunisation coverage (US$40,000), World Health Organization. Research contract — 2024–2025

Social Justice Research Education Fund (US$10,000), Reed College. Deepening Historical Insight: Mapping Population's Racial and Socio-Demographics in Portland, with master artist Michael Stevenson Jr. — 2024

Population-level impact of HIV self-testing distribution (US$24,550), World Health Organization, Global HIV Programme. Research contract — 2023

Granular Data Analysis of HIV Burden (US$50,750), UNAIDS Eastern and Southern Africa. Principal investigator — 2023

Impact evaluation model development (US$25,000), French Development Agency, through Solthis. Consultant — 2018–2019

Fellowships, Scholarships and Awards

Postdoctoral Fellowship (CAD 140,000), Canadian Institutes of Health Research. Declined in favour of the award below — 2023–2025

Outstanding International Postdoctoral Fellowship (CAD 90,000), Fonds de Recherche du Québec Santé — 2022–2023

Applied Population Health Research Fellowship (US$15,000), University of Washington Population Health Initiative and King County Public Health — 2021

Junior Investigator Award (US$2,000), Tuberculosis Research and Treatment Center, University of Washington — 2021

PhD Summer Fellowship (US$25,000), Institute for Disease Modeling, Gates Foundation — 2020

Full PhD Scholarship (US$205,000), Institute for Health Metrics and Evaluation, University of Washington — 2017–2021

Travel awards, Department of Epidemiology and Biostatistics, McGill University (2022); Department of Global Health, University of Washington (2018) — 2018, 2022

Best Master's Thesis Award, MPH in Public Health Research, Paris Saclay University — 2016

Master's Training Award (€9,000), French National Institute of Health and Medical Research (INSERM) — 2015–2016

PRINCEPS Master's Training Award for Interdisciplinary Research (€3,000), Sorbonne Paris Cité — 2015–2016

Training Award for International Research Cooperation (€5,000), French Ministry of Foreign Affairs — 2014

Invited Seminars and Talks

Italian National Institute of Statistics (ISTAT), invited workshop, Rome. Calibrating routine HIV testing data for subnational surveillance — Dec 2026

University of Washington, Spatio-Temporal Analyses Bayes seminar, Department of Biostatistics. Statistical calibration of routine facility testing using probability-survey biomarkers to estimate undiagnosed prevalence — 2026

WorldPop, weekly seminar, University of Southampton. Statistical calibration of routine facility testing using probability-survey biomarkers — 2026

S3RI Research Festival, University of Southampton. From routine data to sample placement: model-based adaptive design and joint estimation of HIV indicators in Mozambique — 2025

PhD Population Studies Masterclass, London School of Economics. From a hunch to a manuscript — 2025

Imperial College London, Infectious Diseases Epidemiology Seminar. Temporal trends and major determinants of antenatal care HIV testing — 2022

Socio-Economic Review conference on Elites, Economy and Society, Berlin. Heirs, corporate aristocrats and 'meritocrats': the social space of top CEOs and Chairmen in India — 2017

Conference Presentations

Royal Statistical Society International Conference, Bournemouth. Why ethnography? Studying AI adoption in official statistics as a problem of scientific practice — 2026

Conference on Small Area Estimation (SAE 2026). Calibrating routine facility testing against probability-survey biomarkers for district-level estimation of undiagnosed HIV prevalence — 2026

IASS–ITACOSM joint conference, Bologna. Mismatched measures: comparing asset-based wealth index and income and consumption poverty in Mozambique — 2025

Population Association of America, Washington DC. Re-assessing the DHS wealth index as a proxy for socio-economic status in low resource settings — 2025

International Conference on AIDS and STIs in Africa (ICASA), Harare. Socio-demographic and geographic disparities in HIV burden and care engagement across African countries — 2023

Health Systems Global, Bogotá. A methodological framework to assess disparities in healthcare quality — 2022

International AIDS Conference (AIDS 2018), Amsterdam. Impact evaluation of an HIV viral load testing intervention in Guinea — 2018

Workshops and Capacity Building

Routine data methods workshop, Mozambique Ministry of Health, Maputo. Second week-long workshop with the monitoring team — Nov 2026

Routine health information systems for HIV surveillance, Mozambique Ministry of Health, Maputo. Week-long workshop for the monitoring and evaluation team, funded by WHO and the British Academy — 2025

UNAIDS regional training workshop on HIV estimates and projections, Abidjan. Facilitator for the HIV estimates teams of 17 Francophone ministries of health — 2023

Teaching

Educational leadership

Undergraduate Studies Lead, Department of Social Statistics and Demography, University of Southampton

- Lead the undergraduate module offer with partner departments across the School and Faculty.
- Programme coordination: quality assurance, annual reporting, external examination management.
- Curriculum and business case for the department's new undergraduate degree; reviving a population module.

Departmental teaching away day, Department of Social Statistics and Demography. Organiser — 2026

Undergraduate

Applied Social Quantitative Analyses (RESM3001), University of Southampton. Module lead and principal instructor — 2024–present
Final-year module built end to end in R and Quarto; redesigned from low to high structure; assessment redesigned for students with access to large language models.

Introduction to Social Data Analytics (RESM2001), University of Southampton. Module lead and principal instructor — 2025–present
Cohort rising from 160 to over 250; led its migration from SPSS to R across lectures, tutorials and assessment.

Data Science (MATH 241), Reed College. Instructor — 2024

Introduction to Statistics and Probability (MATH 141), Reed College. Instructor — 2023

Introduction to Statistics and Probability, Sciences Po Paris. Instructor — 2022

Global Health Economics (GH 345), University of Washington. Teaching assistant, 80 students — 2021

Introduction to Statistics and Econometrics, Université Paris I Panthéon-Sorbonne. Teaching assistant — 2014, 2015

Graduate

Advanced Quantitative Methods for Global Health (GH 556), University of Washington. Teaching fellow, core course of the PhD programme — 2020, 2021

Data Analysis and Programming Tools in Health Sciences (HMS 520), University of Washington. Teaching fellow, MSc in Health Metrics Sciences — 2021

Micro-economic Evaluation of Public Policy, Sciences Po Paris. Teaching assistant, Master in Public Policy — 2016

Teaching development and recognition

Fellow of the Higher Education Academy (FHEA), Advance HE — 2026

Professional Recognition of Education Practice, University of Southampton — 2025

Reed Center for Teaching and Learning, Reed College — 2023–2024
Inclusive Pedagogy Circle (weekly reading of Hogan and Sathy's *Inclusive Teaching*); peer observation programme with a senior colleague; Student Teaching Consultant programme.

Supervision and Mentoring

Doctoral

Clare Mitchell, PhD, University of Southampton. Primary supervisor (60%), with Kristine Nilsen and Joseph Larmarange; funded by the Economic and Social Research Council — 2026–present
The politics of indicators in global health.

Dagim Fisseha Habteyesus, PhD, University of Southampton. Co-supervisor (20%), with Kristine Nilsen and Sarah Neal; funded by the Economic and Social Research Council — 2025–present
Measuring sexual and reproductive health indicators in four refugee camps in Ethiopia.

Master's and undergraduate

MSc and undergraduate dissertations, University of Southampton. Five to ten each year — 2024–present

Senior theses and funded summer research, Reed College. Two senior theses and two summer research projects (Asa Ferguson, Louise Xu, Omar Youssif) — 2023–2024

Global Health Scholars, McGill University. Caroline Hodgins (HIV prevalence among men who paid for sex) and Paul Muset (determinants of HIV testing at antenatal care, co-author of the 2024 JAIDS article) — 2022–2023

UNAIDS Granular Data Analysis, McGill University. Three PhD students — 2023

Professional Experience

World Health Organization, Geneva. Technical consultant, HIV testing team — 2023, 2025–26
Multi-country analysis of HIV self-testing scale-up in eight African countries; HIV self-testing monitoring in Mozambique.

World Health Organization, Geneva. Technical consultant, Health Inequality Monitoring — 2023–2024
Landscaping review of spatial analysis methods for childhood immunisation coverage, with Gavi, UNICEF and WHO Immunization.

UNAIDS Regional Support Team for Eastern and Southern Africa, Johannesburg. Consultant — 2022–2023
Inequalities in HIV outcomes across the region; 18 country reports.

Institute for Disease Modeling, Gates Foundation, Seattle. PhD summer fellow, health systems team — 2020

International Training and Education Center for Health (I-TECH) and Haiti Ministry of Health, Seattle and Port-au-Prince. PhD research associate — 2018–2021
Statistical methods for iSanté, Haiti's national HIV electronic medical record, to monitor adherence to treatment guidelines; implemented as an automated layer of the national dashboard.

Solthis, Paris, Guinea and Sierra Leone. Scientific consultant — 2018–2019
Impact evaluation of an HIV viral load testing intervention in Guinea; R package ImpactSimul; user empowerment in HIV care in Sierra Leone.

Gustave Roussy Institute of Oncology, Villejuif. Biostatistician, Department of Biostatistics and Clinical Trials — 2016–2017

Institut Pasteur, Paris. Master's researcher, Biostatistics, Biomathematics and Pharmaco-epidemiology — 2015–2016

Centre de Sciences Humaines, New Delhi. Research intern, economics and sociology — 2014

Service

Department and university

Southampton Statistical Sciences Research Institute (S3RI), Deputy Director; co-organiser of the annual research festival — 2026–present

Methodological Innovation for Survey Methods and Official Statistics seminar, University of Southampton. Revived and organise the monthly series — 2024–present

Ethics reviewer, Faculty of Social Sciences, University of Southampton — 2024–present

Departmental away day on stakeholder and community engagement, co-led with Professor Ann Berrington — 2025

Curriculum Committee, Department of Global Health, University of Washington. Student representative — 2019–2021

Diversity, Equity and Inclusion Committee, Department of Global Health, University of Washington. Student representative — 2020–2021

Discipline

Royal Statistical Society, Fellow; International Development Section committee — 2025–present

Royal Statistical Society International Conference, Conference Board, 2026 conference; chair of the History of Statistics Section session "Hidden Figures" — 2025–2026

PhD Population Studies Masterclass, co-organised with LSE and the UBEL and South Coast doctoral training partnerships. Co-organiser 2025, lead 2026 — 2025–present

UNAIDS Reference Group on Estimates, Modelling and Projections, member — 2022–2024

WHO technical working group on HIV self-testing, member — 2023

Reviewing and assessment

Economic and Social Research Council, assessor, Advanced Quantitative Methods doctoral studentships — 2026

Fonds de Recherche du Québec Santé, reviewer, master's fellowships — 2023

McGill University, reviewer, undergraduate Global Health Scholarships — 2023

Sciences Po Paris, undergraduate admissions committee — 2022, 2023

Reviewer service

Journal of Official Statistics, *BMJ Global Health*, *BMC Public Health*, *Journal of the International AIDS Society*, *JAIDS Journal of Acquired Immune Deficiency Syndromes*, *The Lancet Global Health*

Volunteering

Chess in prisons, playing chess with prisoners, most recently at HMP Brixton, London

La Cimade, Paris. Legal support for foreigners and migrant workers — 2022–2023

ROOTS Young Adult Shelter, Seattle. Dinner and overnight shifts for young people experiencing homelessness — 2019–2020

Skills and Interests

Computing, primarily R (Stan, INLA, Quarto, Shiny, survey, TMB, tidyverse); also Python, Stata, SAS and SPSS; LaTeX, Git and GitHub

Languages, French (native), English (C2), Portuguese (A2), German (A2)

Chess, France High School Champion (2009); France University Champion (2010)